FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

COMPUTER SCIENCE

Past Paper 2025 • Paper Code: 1HA-I 25007 — 30071 (Version B) / 32071 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

COMPUTER SCIENCE HSSC-I — SECTION – A (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 13 | **Paper Code:** 30071 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Which gate produces a HIGH output when at least one input is LOW?

(A) AND gate
(B) NAND gate
(C) OR gate
(D) NOR gate

Q2. What is an efficient way to organize a collection of photos on your computer?

(A) Naming them in random order
(B) Grouping them by date or event
(C) Deleting some photos randomly
(D) Printing all photos and shuffling them

Q3. Which of the following represents a reliable source of information?

(A) A peer-reviewed academic journal
(B) A social media post without citation
(C) A personal blog with no evidence
(D) A random website with advertisements

Q4. What is the main objective of classification in Supervised learning?

(A) To group data into clusters without labels
(B) To predict continuous values
(C) To reduce the dimensions of data
(D) To assign predefined labels to input data

Q5. Which type of cloud deployment model is most suitable for organizations with strict regulatory requirements?

(A) Private cloud
(B) Public cloud
(C) Hybrid cloud
(D) Community cloud

Q6. Consider the list, P= [2, 5, 3, 4], what will the statement print (max(P)) return?

(A) 2
(B) 4
(C) 5
(D) Error in statement

Q7. What is the primary goal of abstraction in computational thinking?

(A) Writing detailed step–by–step instructions for solving a problem
(B) Using a computer to solve mathematical equations
(C) Repeating a process until a solution is found
(D) Ignoring unnecessary details to focus on the important aspects of problem

Q8. Which type of feasibility study in SDLC involves evaluating whether the proposed system can be developed with available technology?

(A) Legal feasibility
(B) Schedule feasibility
(C) Technical feasibility
(D) Economic feasibility

Q9. Which of the following is an example of open-ended research question?

(A) How many students use mobile phones in a class?
(B) Does exercise reduce stress?
(C) What are the effects of climate change on agriculture?
(D) Is online education better than traditional education?

Q10. Which of the following can be classified as a high-fidelity prototype?

(A) A detailed mobile app demo with clickable buttons
(B) A hand-drawn wireframe of a website
(C) A basic outline of an app on paper
(D) A simple text document describing the app features

Q11. What is the main advantage of using a simulation instead of real-world testing?

(A) Always gives perfect results
(B) Risk-free and cost-effective
(C) Eliminates the need for real-world experience
(D) Requires no computer programming

Q12. What will be the result of x=2 | 4 Python statement?

(A) 0 (B) 2

(C) 4 (D) 6

Q13. Which of the following is an example of a service?

(A) e-book (B) Laptop

(C) Medical consultation (D) Pre-packaged food items

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:40 Hours | **Maximum Marks:** 62 | **Paper Code:** 1HA-I 25007 (B)

Q2. (Marks (14 x 3 = 42)) Q.2 Answer the following parts briefly.

(i) [03] Assume the variables x, y and z where x=4, y=10, and z=2. What value will be stored in result1, result2 and result3 after executing following Python statements? a) result1 = y − x b) result2 = y // z c) result3 = x % z — OR (03) — Trace the following pseudocode and complete the table below, which tracks the values of the variables at each iteration of the loop. SET product =1 FOR i FROM 1 TO 3 DO product = product * i END FOR PRINT product — [A blank trace table with columns: Iteration No. | Value of i | Value of sum | Print is printed in the source scan]

(ii) [03] Write down any three features of data visualization. — OR (1+2) — What is Permissioned Block chain network? Also list down its applications. (any two)

(iii) [03] Write down three comparisons between Symmetric and Asymmetric encryption. — OR (03) — Differentiate between supervised and unsupervised learning with daily life examples.

(iv) [03] Why is prototyping important? Give three reasons. — OR (03) — Write down three differences between Black Box and White Box testing methods.

(v) [03] Determine whether these variable names are valid or invalid. Provide reasons for any invalid variable names. a) input b) your age c) marks1 — OR (03) — Why is two-factor authentication (2FA) important? Also give example.

(vi) [03] Briefly explain three harmful effects of increased connectivity on environment. — OR (03) — Write down three positive impacts of AI systems.

(vii) [1+2] Which algorithm is more efficient: insertion sort or bubble sort? Justify your choice with two reasons. — OR (03) — Correct any three errors in the following Python code: temperature = 5; if temperature > 30: print("It's hot) elif temperature = 5: display("It's cold") else: print("The weather is mild") — [The source scan prints 'print("It's hot)' without a closing quote, and 'elif temperature = 5' with a single equals sign, as shown]

(viii) [03] Evaluate the following expression in Python using the proper order of operations: 2+2**3//2*4%3-1 — OR (03) — Compare Bus and Mesh network topologies with respect to design, scalability and reliability.

(ix) [03] Briefly explain any three guidelines to conduct a better qualitative interview. — OR (03) — Write down any three key steps in developing the Minimum Viable Product.

(x) [03] Write a Python program that asks the user to input name and marks. Also displays the entered name and marks. — OR (03) — Draw truth table for the Boolean function. $F=\bar{X}YZ+X\,\bar{Y}\,Z$

(xi) [03] What is the importance of Digital literacy? Give three reasons. — OR (03) — Write down any three tasks suitable for computing devices.

(xii) [03] Why is horizontal scalability considered more suitable for cloud-based applications? Give three reasons. — OR (03) — What will be generated by the following Python code? country=["Pakistan", "Iran", "China"] country [2]= "Turkey" country.append("Oman") print(country)

(xiii) [03] Compare Parameter and Statistics with daily life example. — OR (03) — Write Python statements considering turtle graphics to perform the following actions: a) Change the pen colour to blue. b) Move the turtle backward by 150 pixels. c) Rotate the turtle 90 degrees to the left.

(xiv) [1+2] What is the deployment phase in SDLC? How does the pilot deployment method differ from the parallel deployment method? — OR (1+2) — What is a sample? Why is it often used instead of studying an entire population?

> Note: SECTION – C (Marks 20) — Attempt the following questions. (4 x 5 = 20)

Q3. (Marks 05) Q.3 Write down any five differences between Waterfall and Agile software development models. — OR (05) — Explain the differences between Correlation and Causation with examples.

Q4. (Marks 05) Q.4 Simplify the Boolean Function F, using Karnaugh Map. Also draw the logic circuit for simplified expression. $F=\bar{A}\bar{B}C+\bar{A}BC+\bar{A}\bar{B}\bar{C}+\bar{A}\bar{B}C+A\bar{B}\bar{C}+ABC$ — OR (05) — Write a Python program that reads two numbers and prints the minimum number by using function.

Q5. (Marks 05) Q.5 Use the Linear Search algorithm to find the number 53 from the given set of numbers: 89 50 36 42 11 53 65 72 24 — OR (05) — Write a Python program that inputs a number and prints its multiplication table from 1 to 10 using for loop.

Q6. (Marks 1+4) Q.6 What is meant by Internet of Things (IoT)? Describe any four applications of IoT technology. — OR (1+4) — What are information sources? Explain any four categories of information sources.

VERSION D

COMPUTER SCIENCE HSSC-I — SECTION – A (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 13 | **Paper Code:** 32071 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Which gate produces a LOW output when at least one input is HIGH?

(A) AND gate
(B) NAND gate
(C) NOR gate
(D) OR gate

Q2. What is an efficient algorithm for finding a specific book in a large library?

(A) Asking other visitors for help
(B) Using the catalog system to find its location
(C) Randomly pulling out books from the shelves
(D) Browsing without a plan

Q3. Which of the following represents a reliable source of information?

(A) A personal blog with no evidence
(B) A social media post without citation
(C) A peer-reviewed academic journal
(D) A random website with advertisements

Q4. K-means clustering is an example of the following statistical modelling technique:

(A) Supervised learning
(B) Semi-supervised learning
(C) Reinforcement learning
(D) Unsupervised learning

Q5. The key advantage of using a private cloud is:

(A) Higher security and control over data
(B) Lower cost compared to public cloud
(C) Accessible to the public
(D) Reduced need for IT infrastructure

Q6. Consider the list, A= [10, 20, 30, 40], what will the statement print (len(A)) return?

(A) 0
(B) 4
(C) 40
(D) Error in statement

Q7. What does decomposition mean in computational thinking?

(A) Combining smaller problems into a larger solution

(B) Ignoring the problem until it resolves itself

(C) Writing code without any structure

(D) Breaking a problem into smaller, more manageable parts

Q8. Which type of feasibility study in SDLC assesses whether the proposed system can be completed within a given timeframe?

(A) Legal feasibility

(B) Schedule feasibility

(C) Technical feasibility

(D) Operational feasibility

Q9. Which of the following is an example of a typical survey question?

(A) Solve 5x7=?

(B) What is the capital of China?

(C) How many hours do you spend on the internet daily?

(D) What is the boiling point of water?

Q10. Which of the following can be classified as a low-fidelity prototype?

(A) A wireframe sketch on paper

(B) A fully functional mobile app

(C) A high-resolution mock up with animations

(D) A completed website with backend functionality

Q11. Which of the following is a characteristic of qualitative interviews?

(A) They have only one-word answers

(B) They focus on open-ended questions

(C) They only collect numerical data

(D) They follow a strict set of yes/no questions

Q12. What will be the result of x = 5 & 3 Python statement?

(A) 1

(B) 3

(C) 5

(D) 7

Q13. Which of the following is an example of a product?

(A) Online banking

(B) Car repair

(C) Mobile phone

(D) Medical consultation

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:40 Hours | **Maximum Marks:** 62 | **Paper Code:** 1HA-I 25007 (D)

Q2. (Marks (14 x 3 = 42)) Q.2 Answer the following parts briefly.

(i) [03] Assume the variables x, y and z where x=4, y=8, and z=2. What value will be stored in result1, result2 and result3 after executing following Python statements? a) result1 = x + y b) result2 = z ** 3 c) result3 = y/x — OR (03) — Trace the following pseudocode and complete the table below, which tracks the values of the variables at each iteration of the

loop: SET sum=0 FOR i FROM 1 TO 3 DO sum = sum + i END FOR PRINT sum — [A blank trace table with columns: Iteration no. | Value of i | Value of sum | Print is printed in the source scan]

(ii) [2+1] What is a use case for statistical modelling? Give an example. — OR (03) — Write down three benefits of integration of Blockchain and IoT.

(iii) [03] How does Agile model differ from Waterfall model? Justify with at least three reasons. — OR (1.5+1.5) — Differentiate between Parameter and Statistics with daily life examples.

(iv) [03] Why is successful product development important? Give three reasons. — OR (03) — Write down three differences between analog and digital signals.

(v) [03] Determine whether these variable names are valid or invalid. Provide reasons for any invalid variable names. a) range b) total marks c) counter — OR (03) — How do computer systems protect against cyber-attacks? Provide three reasons.

(vi) [03] Briefly explain three guidelines regarding the safe and responsible use of information sources. — OR (1+2) — What are embedded systems? Enlist two of their applications.

(vii) [1+2] Which algorithm is more efficient: insertion sort or bubble sort? Justify your choice with two reasons. — OR (03) — Correct any three errors in the following Python code: number = 0; if number > 0: print("Positive number") elif number = 0: print("Zero") else: print("Negative number") — [The source scan prints 'elif number = 0' with a single equals sign, as shown]

(viii) [03] Evaluate the following expression in Python using the proper order of operations: 3**2-8%4+10/5*3 — OR (03) — Compare Star and Ring network topologies with respect to design, scalability and reliability.

(ix) [03] Write down any three features of infographics for data representation. — OR (03) — Write down any three key steps in developing the Minimum Viable Product.

(x) [03] Write a Python program that asks the user to input name and age also displays the entered name and age. — OR (03) — Draw truth table for the Boolean function. F=XYZ+X $\bar{Y}$ Z

(xi) [03] Write down three features of the Survey for data collection. — OR (2+1) — What is meant by data searches? Illustrate with an example.

(xii) [03] How does vertical scalability differ from horizontal scalability in cloud computing? Illustrate with three differences. — OR (03) — What will be generated by the following Python code? list1=[34, "abc", 40, 89] list1[2]=67 list1.append("xyz") print(list1)

(xiii) [03] What is the difference between a population and a sample in statistics? Illustrate with example. — OR (03) — Write Python statements considering turtle graphics to perform the following actions: a) Rotate the turtle 45 degree to the right. b) Move the turtle forward by 100 pixels. c) Change the pen colour to red.

(xiv) [1+2] What is the deployment phase in SDLC? How does the direct deployment method differ from the phased deployment method? — OR (1.5+1.5) — Compare supervised and unsupervised learning with daily life examples.

> Note: SECTION – C (Marks 20) — Attempt the following questions. (4 x 5 = 20)

Q3. (Marks 05) Q.3 What is encryption? Differentiate between symmetric and asymmetric encryption in terms of speed, complexity, security and key structure. — OR (1+4) — What is data collection? Compare primary and secondary data collection methods based on the originality, cost, time required and reliability.

Q4. (Marks 05) Q.4 Using a 3-variable Karnaugh Map, simplify the following Boolean Function F. Also represent the simplified Boolean expression with a logic circuit diagram. $F=A\bar{B}\bar{C}+\bar{A}\bar{B}C+A\bar{B}C+\bar{A}BC+ABC+\bar{A}B\bar{C}$ — OR (05) — Write a Python program that reads three numbers and prints their product by using function.

Q5. (Marks 05) Q.5 Use the Binary Search algorithm to find the number, 75 from the given set of numbers: 9 10 26 34 41 58 67 75 83 — OR (05) — Write a Python program that takes a number and prints its factorial using for loop.

Q6. (Marks 1+4) Q.6 What does Artificial Intelligence mean? Describe any four positive impacts of AI systems. — OR (1+4) — What are assistive technologies? Provide four examples of their everyday uses.